Process Intelligence Platform

Enterprise Entity-Relationship Diagram

Ontologically-Driven Architecture Specification

Aligned Standards & Libraries
PM4Py • OCEL 2.0 • Object-Centric Petri Nets • XES • PQL

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1. Executive Summary

This document presents the complete Entity-Relationship Diagram (ERD) for an enterprise-grade Process Mining and Business Automation SaaS platform. The architecture is designed with an **ontological approach**, organizing entities into distinct layers based on their fundamental nature and purpose within the system.

Key Design Principles

• **Ontological Layering:** Entities are organized by their fundamental nature (existence, identity, commercial, etc.) rather than just functional grouping.

• **OCEL 2.0 Native:** First-class support for Object-Centric Event Logs with direct mapping to PM4Py's OCEL implementation.

• **Dual Paradigm:** Supports both traditional case-centric process mining (XES) and modern object-centric process mining (OCEL).

• **Multi-Tenancy:** Complete tenant isolation with hierarchical organization support.

• **Extensibility:** JSONB metadata fields and schema versioning for future evolution.

• **Audit & Compliance:** Immutable event stores and comprehensive change tracking.

Metric	Count
Ontological Layers	13
Total Entities	40
OCPM-Specific Entities	8
Case-Centric Entities	4
Automation Entities	5

2. Ontological Architecture

The platform's data model is organized into 13 ontological layers, each representing a distinct aspect of "being" within the system. This approach ensures clear separation of concerns and enables modular evolution of the platform.

Layer	Purpose	PM4Py Alignment
Existence Layer	Establishes WHO exists in the system and the organizational boundaries that part...	Tenant isolation for event logs, data models, and ...
Identity Layer	Defines WHO can act within the system and WHAT actions they are permitted to per...	Resource attribute in event logs, user-based fil...
Commercial Layer	Manages the value exchange relationship between platform and customer.	Usage limits for process discovery, conformance ch...
Operational Layer	Controls HOW the system behaves for each tenant and user.	Algorithm parameters, discovery settings, visualiz...
Temporal Layer	Maintains WHEN things happened and preserves historical truth.	Event log immutability, change tracking, complianc...
Integration Layer	Enables data flow FROM and TO external systems.	Event log import (XES, OCEL, CSV), SAP/ERP connect
Communication Layer	Facilitates information flow TO users about system events.	Process deviation alerts, SLA breach notifications...
Data Infrastructure Layer	The physical storage and movement of raw process data.	Event log storage, data model schemas, extraction ...
OCPM Core Layer	Native support for multi-object process mining paradigm.	Direct mapping to pm4py OCEL, Object-Centric Petri...
Case-Centric Layer	Classic XES-aligned process mining constructs.	pm4py EventLog, Trace, Event, Variant objects
Semantic Layer	Translates raw data into business-understandable concepts.	Custom metrics, PQL queries, process perspectives
Studio/UI Layer	User-facing dashboards, process explorers, and analytics.	Process model visualization, variant explorer, KPI...
Automation Layer	Process automation, issue detection, and remediation actions.	Process enhancement, automated conformance monitor

3. Existence Layer

Existence Layer

Root identity & multi-tenancy boundary. The foundational 'being' of the platform.

Purpose: Establishes WHO exists in the system and the organizational boundaries that partition all data.

PM4Py Alignment: Tenant isolation for event logs, data models, and OCEL stores

Tenant — Type: Core

Root organizational boundary and multi-tenancy partition key

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`external_id`	VARCHAR(50)	UNIQUE	External system reference
`name`	VARCHAR(255)	NOT NULL	Organization name
`slug`	VARCHAR(100)	UNIQUE	URL-safe identifier
`type`	ENUM	NOT NULL	individual|team|enterprise
`status`	ENUM	NOT NULL	provisioning|active|suspended|terminated
`tier`	ENUM	NOT NULL	free|starter|professional|enterprise
`metadata`	JSONB	DEFAULT '{}'	Extensible attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp
`updated_at`	TIMESTAMPTZ	NOT NULL	Last update timestamp
`deleted_at`	TIMESTAMPTZ	NULL	Soft delete timestamp

Relationships: 1:N → Organization | 1:N → User | 1:N → DataPool | 1:N → Subscription | 1:1 → TenantSettings

Organization — Type: Core

Hierarchical organizational unit within a tenant (for enterprise)

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`parent_org_id`	UUID	FK→Organization	Parent org (self-ref)
`name`	VARCHAR(255)	NOT NULL	Organization name
`code`	VARCHAR(50)	NOT NULL	Short code
`type`	ENUM	NOT NULL	division|department|unit|custom
`hierarchy_path`	LTREE	NOT NULL	Materialized path for tree queries
`metadata`	JSONB	DEFAULT '{}'	Custom attributes
`is_active`	BOOLEAN	DEFAULT true	Active status
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:1 → Organization | 1:N → Organization

Environment — Type: Core

Isolated runtime context (production, staging, development)

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`name`	VARCHAR(100)	NOT NULL	Environment name
`type`	ENUM	NOT NULL	production|staging|development|sandbox
`is_default`	BOOLEAN	DEFAULT false	Default environment flag
`config`	JSONB	DEFAULT '{}'	Environment configuration
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | 1:N → ApiKey

4. Identity Layer

Identity Layer

Authentication, authorization, and actor management.

Purpose: Defines WHO can act within the system and WHAT actions they are permitted to perform.

PM4Py Alignment: Resource attribution in event logs, user-based filtering

User — Type: Core

Human or service actor in the system *[OCEL: Resource in OCEL events]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`external_id`	VARCHAR(255)	NULL	External IdP reference
`email`	VARCHAR(320)	NOT NULL	Email address
`email_verified_at`	TIMESTAMPTZ	NULL	Email verification time
`phone`	VARCHAR(20)	NULL	Phone number
`password_hash`	VARCHAR(255)	NULL	Hashed password
`status`	ENUM	NOT NULL	pending|active|suspended|deactivated
`type`	ENUM	NOT NULL	human|service|bot
`last_login_at`	TIMESTAMPTZ	NULL	Last login time
`mfa_enabled`	BOOLEAN	DEFAULT false	MFA status
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp
`deleted_at`	TIMESTAMPTZ	NULL	Soft delete timestamp

Relationships: N:1 → Tenant | 1:1 → UserProfile | 1:N → Session | N:M → Role | N:M → Team

Role — Type: Core

Named collection of permissions for RBAC

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	NULL for system roles
`name`	VARCHAR(100)	NOT NULL	Role name
`slug`	VARCHAR(100)	NOT NULL	URL-safe identifier
`description`	TEXT	NULL	Role description
`type`	ENUM	NOT NULL	system|custom
`scope`	ENUM	NOT NULL	global|organization|team|resource
`is_default`	BOOLEAN	DEFAULT false	Default assignment flag
`metadata`	JSONB	DEFAULT '{}'	Custom attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:M → Permission | N:M → User

Permission — Type: Reference

Atomic authorization unit (resource + action)

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`resource`	VARCHAR(100)	NOT NULL	Target resource type
`action`	VARCHAR(50)	NOT NULL	Allowed action
`description`	TEXT	NULL	Permission description
`category`	VARCHAR(50)	NOT NULL	Grouping category
`is_sensitive`	BOOLEAN	DEFAULT false	Sensitive permission flag

Relationships: N:M → Role

Team — Type: Core

Collaborative group of users

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`organization_id`	UUID	FK→Organization	Optional org scope
`name`	VARCHAR(100)	NOT NULL	Team name
`slug`	VARCHAR(100)	NOT NULL	URL-safe identifier
`description`	TEXT	NULL	Team description
`visibility`	ENUM	NOT NULL	private|internal|public
`metadata`	JSONB	DEFAULT '{}'	Custom attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant \| N:1 → Organization \| N:M → User

IdentityProvider

Type: Core

External SSO/IdP configuration

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`name`	VARCHAR(100)	NOT NULL	Provider name
`type`	ENUM	NOT NULL	saml|oidc|ldap|oauth2
`is_enabled`	BOOLEAN	DEFAULT true	Active status
`is_default`	BOOLEAN	DEFAULT false	Default provider
`config_encrypted`	TEXT	NOT NULL	Encrypted configuration
`metadata_url`	TEXT	NULL	SAML metadata URL
`domain_hints`	VARCHAR[]	DEFAULT '{}'	Email domain hints
`auto_provision`	BOOLEAN	DEFAULT true	Auto-create users
`default_role_id`	UUID	FK→Role	Default role for new users
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:1 → Role

5. Data Infrastructure Layer

Data Infrastructure Layer

Data pools, connections, tables, columns, and ETL pipelines.

Purpose: The physical storage and movement of raw process data.

PM4Py Alignment: Event log storage, data model schemas, extraction jobs

DataPool

Type: Core

Container for data connections and tables *[OCEL: OCEL log container]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
name	VARCHAR(255)	NOT NULL	Pool name
description	TEXT	NULL	Pool description
storage_type	ENUM	NOT NULL	internal|external
external_connection_id	UUID	FK→DataConnection	External storage link
schema_name	VARCHAR(100)	NULL	Database schema
storage_bytes	BIGINT	DEFAULT 0	Storage usage
table_count	INTEGER	DEFAULT 0	Table count
status	ENUM	NOT NULL	active|inactive|archived
created_by	UUID	FK→User	Creator
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | 1:N → Table | 1:N → DataModel | 1:N → DataConnection

DataConnection

Type: Core

Configuration for external data source connection *[OCEL: XES/OCEL import source]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_pool_id`	UUID	FK→DataPool	Parent pool
`name`	VARCHAR(255)	NOT NULL	Connection name
`connector_type`	ENUM	NOT NULL	jdbc|rest|sap_rfc|sftp|s3|kafka|salesfor...
`connection_config`	JSONB	NOT NULL	Connection parameters (encrypted fields)
`auth_type`	ENUM	NOT NULL	basic|oauth2|api_key|sap_user|none
`test_query`	TEXT	NULL	Validation query
`last_tested_at`	TIMESTAMPTZ	NULL	Last test timestamp
`last_test_status`	ENUM	NULL	success|failed|pending
`status`	ENUM	NOT NULL	active|invalid|expired|disabled
`created_by`	UUID	FK→User	Creator
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:1 → DataPool | 1:N → DataJob

DataModel

Type: Core

Analytical schema for process mining *[OCEL: OCEL schema definition]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_pool_id`	UUID	FK→DataPool	Parent pool
`name`	VARCHAR(255)	NOT NULL	Model name
`description`	TEXT	NULL	Model description
`model_type`	ENUM	NOT NULL	case_centric|object_centric
`activity_table_id`	UUID	FK→Table	Activity table
`case_table_id`	UUID	FK→Table	Case table
`case_column`	VARCHAR(255)	NULL	Case ID column
`activity_column`	VARCHAR(255)	NULL	Activity column
`timestamp_column`	VARCHAR(255)	NULL	Timestamp column
`sorting_column`	VARCHAR(255)	NULL	Tiebreaker column
`load_status`	ENUM	NOT NULL	pending|loading|loaded|failed|stale
`last_loaded_at`	TIMESTAMPTZ	NULL	Last load timestamp
`version`	INTEGER	DEFAULT 1	Schema version
`...`	...	...	(+2 more attributes)

Relationships: N:1 → Tenant | N:1 → DataPool | 1:N → Perspective | 1:N → KnowledgeModel

Table Type: Core

Physical or virtual table in data pool *[OCEL: OCEL object/event type source]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_pool_id`	UUID	FK→DataPool	Parent pool
`name`	VARCHAR(255)	NOT NULL	Table name
`display_name`	VARCHAR(255)	NULL	Display name
`table_type`	ENUM	NOT NULL	physical|view|materialized|external
`source_connection_id`	UUID	FK→DataConnection	Source connection
`source_schema`	VARCHAR(255)	NULL	Source schema
`source_table`	VARCHAR(255)	NULL	Source table
`transformation_sql`	TEXT	NULL	View SQL
`row_count`	BIGINT	DEFAULT 0	Row count
`size_bytes`	BIGINT	DEFAULT 0	Storage size
`last_sync_at`	TIMESTAMPTZ	NULL	Last sync timestamp
`is_activity_table`	BOOLEAN	DEFAULT false	Activity table flag
`is_case_table`	BOOLEAN	DEFAULT false	Case table flag
`...`	...	...	(+1 more attributes)

Relationships: N:1 → Tenant | N:1 → DataPool | 1:N → Column

Column Type: Supporting

Column definition within a table *[OCEL: OCEL attribute definition]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`table_id`	UUID	FK→Table	Parent table
`name`	VARCHAR(255)	NOT NULL	Column name
`display_name`	VARCHAR(255)	NULL	Display name
`data_type`	ENUM	NOT NULL	string|integer|decimal|boolean|date|date...
`source_data_type`	VARCHAR(100)	NULL	Original data type
`is_nullable`	BOOLEAN	DEFAULT true	Nullable flag
`is_primary_key`	BOOLEAN	DEFAULT false	PK flag
`is_indexed`	BOOLEAN	DEFAULT false	Index flag
`ordinal_position`	INTEGER	NOT NULL	Column order
`description`	TEXT	NULL	Column description
`statistics`	JSONB	DEFAULT '{}'	Column statistics
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Table

DataJob

Type: Core

Orchestrated extraction/transformation workflow *[OCEL: OCEL log import job]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_pool_id`	UUID	FK→DataPool	Parent pool
`name`	VARCHAR(255)	NOT NULL	Job name
`job_type`	ENUM	NOT NULL	extraction|transformation|data_model_loa...
`schedule_id`	UUID	FK→Schedule	Schedule reference
`timeout_seconds`	INTEGER	DEFAULT 3600	Timeout
`retry_count`	INTEGER	DEFAULT 3	Retry attempts
`status`	ENUM	NOT NULL	active|disabled|archived
`last_run_at`	TIMESTAMPTZ	NULL	Last execution
`last_run_status`	ENUM	NULL	pending|running|success|failed|cancelled
`next_run_at`	TIMESTAMPTZ	NULL	Next scheduled run
`created_by`	UUID	FK→User	Creator
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:1 → DataPool | N:1 → Schedule | 1:N → JobExecution

6. OCPM Core Layer

OCPM Core Layer

Object-Centric Process Mining entities aligned with OCEL 2.0 standard.

Purpose: Native support for multi-object process mining paradigm.

PM4Py Alignment: Direct mapping to pm4py OCEL, Object-Centric Petri Nets

Perspective — Type: Core (OCPM)

Filtered view of OCPM data for specific analysis *[OCEL: OCEL log view/filter]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_model_id`	UUID	FK→DataModel	Parent data model
`name`	VARCHAR(255)	NOT NULL	Perspective name
`description`	TEXT	NULL	Description
`included_object_types`	UUID[]	DEFAULT '{}'	Included ObjectTypes
`included_event_types`	UUID[]	DEFAULT '{}'	Included EventTypes
`included_relationships`	UUID[]	DEFAULT '{}'	Included Relationships
`filter_expression`	TEXT	NULL	PQL filter
`is_default`	BOOLEAN	DEFAULT false	Default perspective
`status`	ENUM	NOT NULL	draft|published|deprecated
`created_by`	UUID	FK→User	Creator
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → DataModel | 1:N → ObjectType | 1:N → EventType

ObjectType

Type: Core (OCPM)

Classification of business objects (e.g., Order, Invoice, Customer) *[OCEL: ocel:object-type]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
perspective_id	UUID	FK→Perspective	Parent perspective
name	VARCHAR(100)	NOT NULL	Type name (snake_case)
display_name	VARCHAR(255)	NULL	Display name
description	TEXT	NULL	Description
source_table	VARCHAR(255)	NULL	Source table
identifier_columns	TEXT[]	NOT NULL	Columns forming unique key
icon	VARCHAR(50)	NULL	Icon identifier
color	VARCHAR(7)	NULL	Hex color code
is_lead_object	BOOLEAN	DEFAULT false	Case key for flattening
status	ENUM	NOT NULL	draft|published|deprecated
attribute_schema	JSONB	DEFAULT '{}'	JSON Schema for attributes
created_by	UUID	FK→User	Creator
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Perspective | 1:N → Object | 1:N → ObjectRelationship | 1:N → ObjectRelationship

EventType

Type: Core (OCPM)

Classification of event occurrences in OCPM *[OCEL: ocel:event-type]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
perspective_id	UUID	FK→Perspective	Parent perspective
name	VARCHAR(100)	NOT NULL	Type name
display_name	VARCHAR(255)	NULL	Display name
description	TEXT	NULL	Description
source_table	VARCHAR(255)	NULL	Source table
timestamp_column	VARCHAR(255)	NOT NULL	Timestamp column
sorting_column	VARCHAR(255)	NULL	Tiebreaker column
icon	VARCHAR(50)	NULL	Icon identifier
color	VARCHAR(7)	NULL	Hex color code
status	ENUM	NOT NULL	draft|published|deprecated
attribute_schema	JSONB	DEFAULT '{}'	JSON Schema for attributes
created_by	UUID	FK→User	Creator
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Perspective | 1:N → Event

Object

Type: Core (OCPM) - High Volume

Instance of an ObjectType (e.g., a specific SalesOrder) *[OCEL: ocel:object]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`object_type_id`	UUID	FK→ObjectType	Object type
`object_key`	VARCHAR(255)	NOT NULL	Business identifier
`source_system`	VARCHAR(100)	NULL	Source system
`source_id`	VARCHAR(255)	NULL	Source system ID
`lifecycle_state`	VARCHAR(50)	DEFAULT 'active'	Current state
`first_event_at`	TIMESTAMPTZ	COMPUTED	First event timestamp
`last_event_at`	TIMESTAMPTZ	COMPUTED	Last event timestamp
`event_count`	INTEGER	COMPUTED	Event count
`attributes`	JSONB	DEFAULT '{}'	Object attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → ObjectType | N:M → Event | N:M → Object

ObjectRelationship

Type: Core (OCPM)

Relationship definition between two ObjectTypes *[OCEL: ocel:object-type relationship]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`perspective_id`	UUID	FK→Perspective	Parent perspective
`name`	VARCHAR(100)	NOT NULL	Relationship name
`source_object_type_id`	UUID	FK→ObjectType	Source type
`target_object_type_id`	UUID	FK→ObjectType	Target type
`cardinality`	ENUM	NOT NULL	one_to_one|one_to_many|many_to_one|many_...
`join_columns`	JSONB	NOT NULL	Column mapping
`is_embedded`	BOOLEAN	DEFAULT false	Breaks cycles
`display_name`	VARCHAR(255)	NULL	Display name
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Perspective | N:1 → ObjectType | N:1 → ObjectType | 1:N → ObjectRelationshipInstance

ObjectRelationshipInstance — Type: Junction (OCPM) - High Volume

Instance linking two Objects *[OCEL: ocel:o2o relationship]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`relationship_id`	UUID	FK→ObjectRelationship	Relationship definition
`source_object_id`	UUID	FK→Object	Source object
`target_object_id`	UUID	FK→Object	Target object
`valid_from`	TIMESTAMPTZ	NULL	Validity start
`valid_to`	TIMESTAMPTZ	NULL	Validity end
`attributes`	JSONB	DEFAULT '{}'	Relationship attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → ObjectRelationship | N:1 → Object | N:1 → Object

EventObjectRelationship — Type: Junction (OCPM) - High Volume

Links Events to Objects (N:M in OCPM) *[OCEL: ocel:e2o relationship]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`event_id`	UUID	FK→Event	Event reference
`object_id`	UUID	FK→Object	Object reference
`qualifier`	VARCHAR(100)	NULL	Role of object in event
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Event | N:1 → Object

ObjectChange — Type: Transactional (OCPM) - Immutable

Tracks attribute changes on Objects over time *[OCEL: ocel:object-change]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`object_id`	UUID	FK→Object	Changed object
`attribute_name`	VARCHAR(255)	NOT NULL	Changed attribute
`old_value`	JSONB	NULL	Previous value
`new_value`	JSONB	NULL	New value
`changed_at`	TIMESTAMPTZ	NOT NULL	Change timestamp
`changed_by`	UUID	FK→User	User who made change
`source_event_id`	UUID	FK→Event	Triggering event
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Object | N:1 → Event

7. Case-Centric Layer

Case-Centric Layer

Traditional process mining with cases, events, activities, variants.

Purpose: Classic XES-aligned process mining constructs.

PM4Py Alignment: pm4py EventLog, Trace, Event, Variant objects

Activity — Type: Reference

Named action/step that can occur within a process *[OCEL: pm4py Activity concept]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`name`	VARCHAR(255)	NOT NULL	Activity name
`display_name`	VARCHAR(255)	NULL	Display name
`description`	TEXT	NULL	Description
`category`	VARCHAR(100)	NULL	Category
`icon`	VARCHAR(50)	NULL	Icon
`color`	VARCHAR(7)	NULL	Color
`is_milestone`	BOOLEAN	DEFAULT false	Milestone flag
`is_automated`	BOOLEAN	DEFAULT false	Automated flag
`avg_duration_seconds`	INTEGER	COMPUTED	Average duration
`metadata`	JSONB	DEFAULT '{}'	Custom attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | 1:N → Event

Case

Type: Core

Single execution instance of a process (case-centric view) *[OCEL: pm4py Trace / XES case]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_model_id`	UUID	FK→DataModel	Parent data model
`case_key`	VARCHAR(255)	NOT NULL	Business case ID
`source_system_id`	VARCHAR(100)	NULL	Source system
`started_at`	TIMESTAMPTZ	COMPUTED	Start time
`ended_at`	TIMESTAMPTZ	COMPUTED	End time
`throughput_time_seconds`	BIGINT	COMPUTED	Throughput time
`event_count`	INTEGER	COMPUTED	Event count
`variant_id`	UUID	FK→Variant	Variant reference
`is_complete`	BOOLEAN	COMPUTED	Completion flag
`attributes`	JSONB	DEFAULT '{}'	Case attributes
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:1 → DataModel | N:1 → Variant | 1:N → Event

Event

Type: Core - High Volume - Immutable

Single occurrence of an activity with timestamp *[OCEL: pm4py Event / XES event / ocel:event]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`case_id`	UUID	FK→Case	NULL for OCPM-only
`activity_id`	UUID	FK→Activity	Activity reference
`activity_name`	VARCHAR(255)	NOT NULL	Denormalized activity name
`timestamp`	TIMESTAMPTZ	NOT NULL	Event timestamp
`sorting_key`	INTEGER	NULL	Tiebreaker
`resource_id`	UUID	FK→User	Resource reference
`resource_name`	VARCHAR(255)	NULL	Denormalized resource
`lifecycle_state`	ENUM	DEFAULT 'complete'	schedule|start|complete|suspend|resume|a...
`source_system`	VARCHAR(100)	NULL	Source system
`source_table`	VARCHAR(255)	NULL	Source table
`source_id`	VARCHAR(255)	NULL	Source ID
`cost`	DECIMAL(18,4)	NULL	Event cost
`duration_seconds`	INTEGER	NULL	Duration
`...`	...	...	(+2 more attributes)

Relationships: N:1 → Tenant | N:1 → Case | N:1 → Activity | N:M → Object

Variant

Type: Computed - Materialized

Unique sequence of activities representing a process path *[OCEL: pm4py Variant]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`data_model_id`	UUID	FK→DataModel	Parent data model
`activity_sequence`	TEXT[]	NOT NULL	Ordered activity names
`sequence_hash`	VARCHAR(64)	NOT NULL	SHA256 hash
`case_count`	INTEGER	COMPUTED	Number of cases
`percentage`	DECIMAL(5,2)	COMPUTED	Percentage of total
`avg_throughput_seconds`	BIGINT	COMPUTED	Avg throughput
`is_happy_path`	BOOLEAN	DEFAULT false	Happy path flag
`rank`	INTEGER	COMPUTED	Rank by frequency
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | N:1 → DataModel | 1:N → Case

8. Semantic Layer

Semantic Layer

Knowledge models, KPIs, records, filters, and business meaning.

Purpose: Translates raw data into business-understandable concepts.

PM4Py Alignment: Custom metrics, PQL queries, process perspectives

KnowledgeModel

Type: Core

Semantic layer containing KPIs, Records, Filters, Variables *[OCEL: Business abstraction layer]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`package_id`	UUID	FK→Package	Parent package
`key`	VARCHAR(100)	NOT NULL	Unique key (snake_case)
`name`	VARCHAR(255)	NOT NULL	Display name
`description`	TEXT	NULL	Description
`data_model_id`	UUID	FK→DataModel	Linked data model
`km_type`	ENUM	NOT NULL	base|extension
`extends_km_id`	UUID	FK→KnowledgeModel	Extended KM
`status`	ENUM	NOT NULL	draft|published
`version`	INTEGER	DEFAULT 1	Version number
`yaml_content`	TEXT	NULL	YAML source
`published_at`	TIMESTAMPTZ	NULL	Publish timestamp
`created_by`	UUID	FK→User	Creator
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Package | N:1 → DataModel | 1:N → KPI | 1:N → Record | 1:N → Filter

KPI

Type: Core

Calculated metric with PQL formula *[OCEL: Process metric]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
knowledge_model_id	UUID	FK→KnowledgeModel	Parent KM
key	VARCHAR(100)	NOT NULL	Unique key
display_name	VARCHAR(255)	NULL	Display name
description	TEXT	NULL	Description
pql_expression	TEXT	NOT NULL	PQL formula
return_type	ENUM	NOT NULL	number|string|date|boolean|array
format_string	VARCHAR(50)	NULL	D3 format
unit	VARCHAR(50)	NULL	Unit of measure
aggregation_type	ENUM	NOT NULL	sum|avg|min|max|count|count_distinct|cus...
is_global	BOOLEAN	DEFAULT false	Global scope
category	VARCHAR(100)	NULL	Category
thresholds	JSONB	DEFAULT '[]'	Color thresholds
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → KnowledgeModel

Record

Type: Core

Abstraction of DataModel table with attributes *[OCEL: Business entity view]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
knowledge_model_id	UUID	FK→KnowledgeModel	Parent KM
key	VARCHAR(100)	NOT NULL	Unique key
display_name	VARCHAR(255)	NULL	Display name
description	TEXT	NULL	Description
base_table	VARCHAR(255)	NOT NULL	Source table
identifier_attribute	VARCHAR(255)	NULL	ID attribute
default_sort_attribute	VARCHAR(255)	NULL	Default sort
default_sort_order	ENUM	DEFAULT 'asc'	asc|desc
icon	VARCHAR(50)	NULL	Icon
color	VARCHAR(7)	NULL	Color
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → KnowledgeModel | 1:N → RecordAttribute

Filter

Type: Core

Reusable PQL-based filter condition *[OCEL: Event log filter]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`knowledge_model_id`	UUID	FK→KnowledgeModel	Parent KM
`key`	VARCHAR(100)	NOT NULL	Unique key
`display_name`	VARCHAR(255)	NULL	Display name
`description`	TEXT	NULL	Description
`pql_expression`	TEXT	NOT NULL	PQL condition
`base_table`	VARCHAR(255)	NULL	Base table
`filter_type`	ENUM	NOT NULL	standard|process|forced
`category`	VARCHAR(100)	NULL	Category
`is_default`	BOOLEAN	DEFAULT false	Default filter
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → KnowledgeModel

EventLogConfig

Type: Core

Configuration mapping data to process mining format *[OCEL: XES/OCEL mapping config]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`knowledge_model_id`	UUID	FK→KnowledgeModel	Parent KM
`key`	VARCHAR(100)	NOT NULL	Unique key
`display_name`	VARCHAR(255)	NULL	Display name
`activity_table`	VARCHAR(255)	NOT NULL	Activity table
`case_id_column`	VARCHAR(255)	NOT NULL	Case ID column
`activity_column`	VARCHAR(255)	NOT NULL	Activity column
`timestamp_column`	VARCHAR(255)	NOT NULL	Timestamp column
`sorting_column`	VARCHAR(255)	NULL	Tiebreaker column
`resource_column`	VARCHAR(255)	NULL	Resource column
`cost_column`	VARCHAR(255)	NULL	Cost column
`included_activities`	TEXT[]	DEFAULT '{}'	Included activities
`excluded_activities`	TEXT[]	DEFAULT '{}'	Excluded activities
`filter_expression`	TEXT	NULL	Filter
`...`	...	...	(+2 more attributes)

Relationships: N:1 → KnowledgeModel

9. Studio/UI Layer

Studio/UI Layer

Spaces, packages, views, components, and visual analysis.

Purpose: User-facing dashboards, process explorers, and analytics.

PM4Py Alignment: Process model visualization, variant explorer, KPI dashboards

Space

Type: Core

Organizational container for packages

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`name`	VARCHAR(255)	NOT NULL	Space name
`description`	TEXT	NULL	Description
`icon`	VARCHAR(50)	DEFAULT 'folder'	Icon
`color`	VARCHAR(7)	DEFAULT '#1890ff'	Color
`is_default`	BOOLEAN	DEFAULT false	Default space
`status`	ENUM	NOT NULL	active|archived
`sort_order`	INTEGER	DEFAULT 0	Sort order
`created_by`	UUID	FK→User	Creator
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Tenant | 1:N → Package

Package

Type: Core

Container for Studio assets (Views, KMs, ActionFlows)

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
space_id	UUID	FK→Space	Parent space
key	VARCHAR(100)	NOT NULL	Unique key
name	VARCHAR(255)	NOT NULL	Package name
description	TEXT	NULL	Description
icon	VARCHAR(50)	DEFAULT 'box'	Icon
color	VARCHAR(7)	DEFAULT '#1890ff'	Color
status	ENUM	NOT NULL	draft|published|archived
version	VARCHAR(20)	DEFAULT '1.0.0'	Version
is_template	BOOLEAN	DEFAULT false	Template flag
source_package_id	UUID	FK→Package	Source template
published_at	TIMESTAMPTZ	NULL	Publish timestamp
created_by	UUID	FK→User	Creator
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Space | 1:N → View | 1:N → KnowledgeModel | 1:N → ActionFlow

View

Type: Core

Interactive dashboard/application *[OCEL: Process visualization]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
package_id	UUID	FK→Package	Parent package
key	VARCHAR(100)	NOT NULL	Unique key
name	VARCHAR(255)	NOT NULL	View name
description	TEXT	NULL	Description
knowledge_model_id	UUID	FK→KnowledgeModel	Linked KM
view_type	ENUM	NOT NULL	standard|profile|extension
layout_mode	ENUM	NOT NULL	scale_to_fit|custom_height
layout_config	JSONB	DEFAULT '{}'	Grid layout
icon	VARCHAR(50)	DEFAULT 'layout'	Icon
is_home	BOOLEAN	DEFAULT false	Home view
sort_order	INTEGER	DEFAULT 0	Sort order
status	ENUM	NOT NULL	draft|published
created_by	UUID	FK→User	Creator
...	...	...	(+1 more attributes)

Relationships: N:1 → Package | N:1 → KnowledgeModel | 1:N → Component

Component

Type: Supporting

UI component within a View *[OCEL: Visualization widget]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
view_id	UUID	FK→View	Parent view
parent_id	UUID	FK→Component	Parent component
component_type	ENUM	NOT NULL	chart|table|kpi_list|process_explorer|va...
key	VARCHAR(100)	NOT NULL	Unique key
title	VARCHAR(255)	NULL	Title
layout_position	JSONB	NOT NULL	x, y, width, height
data_config	JSONB	DEFAULT '{}'	PQL queries, record refs
visual_config	JSONB	DEFAULT '{}'	Colors, formats
interaction_config	JSONB	DEFAULT '{}'	Filters, selections
is_visible	BOOLEAN	DEFAULT true	Visibility
sort_order	INTEGER	DEFAULT 0	Sort order
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → View | N:1 → Component | 1:N → Component

10. Automation Layer

Automation Layer

Action flows, skills, sensors, signals, tasks, and workflows.

Purpose: Process automation, issue detection, and remediation actions.

PM4Py Alignment: Process enhancement, automated conformance monitoring

ActionFlow

Type: Core

Visual automation workflow *[OCEL: Process automation]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`package_id`	UUID	FK→Package	Parent package
`key`	VARCHAR(100)	NOT NULL	Unique key
`name`	VARCHAR(255)	NOT NULL	Flow name
`description`	TEXT	NULL	Description
`trigger_type`	ENUM	NOT NULL	manual|scheduled|event|webhook|sensor
`schedule_id`	UUID	FK→Schedule	Schedule reference
`webhook_id`	UUID	FK→Webhook	Webhook reference
`sensor_id`	UUID	FK→Sensor	Sensor reference
`status`	ENUM	NOT NULL	draft|active|inactive|deactivated
`timeout_seconds`	INTEGER	DEFAULT 3600	Timeout
`blueprint`	JSONB	NOT NULL	Module graph definition
`inputs`	JSONB	DEFAULT '[]'	Input parameters
`outputs`	JSONB	DEFAULT '[]'	Output parameters
`...`	...	...	(+3 more attributes)

Relationships: N:1 → Package | 1:N → ActionFlowModule | 1:N → ActionFlowExecution

Sensor

Type: Core

Detector that identifies data conditions and creates Signals *[OCEL: Conformance monitoring]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`knowledge_model_id`	UUID	FK→KnowledgeModel	Parent KM
`name`	VARCHAR(255)	NOT NULL	Sensor name
`description`	TEXT	NULL	Description
`sensor_type`	ENUM	NOT NULL	record_based|data_model_based|ml_based
`record_id`	UUID	FK→Record	Record reference
`filter_id`	UUID	FK→Filter	Filter reference
`filter_expression`	TEXT	NULL	PQL filter
`identifier_columns`	TEXT[]	DEFAULT '{}'	ID columns
`evaluation_trigger`	ENUM	NOT NULL	data_model_reload|km_publish|scheduled|m...
`max_signals_per_evaluation`	INTEGER	DEFAULT 10000	Max signals
`status`	ENUM	NOT NULL	active|inactive
`last_evaluated_at`	TIMESTAMPTZ	NULL	Last evaluation
`created_by`	UUID	FK→User	Creator
...	...	...	(+1 more attributes)

Relationships: N:1 → KnowledgeModel | 1:N → Signal | 1:N → Skill

Signal

Type: Transactional

Detected data incident from Sensor *[OCEL: Process deviation]*

Attribute	Type	Constraint	Description
`id`	UUID	PK	Primary key
`tenant_id`	UUID	FK→Tenant	Parent tenant
`sensor_id`	UUID	FK→Sensor	Source sensor
`skill_id`	UUID	FK→Skill	Related skill
`record_key`	VARCHAR(255)	NOT NULL	Record identifier
`signal_data`	JSONB	NOT NULL	Detection data
`status`	ENUM	NOT NULL	open|in_progress|snoozed|resolved|dismis...
`priority`	ENUM	NOT NULL	critical|high|medium|low
`assignee_id`	UUID	FK→User	Assigned user
`resolved_at`	TIMESTAMPTZ	NULL	Resolution time
`resolved_by`	UUID	FK→User	Resolver
`task_id`	UUID	FK→Task	Created task
`detected_at`	TIMESTAMPTZ	NOT NULL	Detection time
`created_at`	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Sensor | N:1 → User | 1:1 → Task

Task

Type: Core

Work item assigned to users *[OCEL: Work management]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
task_type_id	UUID	FK→TaskType	Task type
title	VARCHAR(255)	NOT NULL	Task title
description	TEXT	NULL	Description
status	ENUM	NOT NULL	open|in_progress|resolved|cancelled
priority	ENUM	NOT NULL	critical|high|medium|low
assignee_id	UUID	FK→User	Assigned user
reporter_id	UUID	FK→User	Reporter
due_date	DATE	NULL	Due date
completed_at	TIMESTAMPTZ	NULL	Completion time
related_signal_id	UUID	FK→Signal	Source signal
related_record_type	VARCHAR(100)	NULL	Related record type
related_record_key	VARCHAR(255)	NULL	Related record key
attributes	JSONB	DEFAULT '{}'	Custom attributes
...	...	...	(+1 more attributes)

Relationships: N:1 → User | N:1 → Signal

Skill

Type: Core

Reusable automation combining Sensor + Actions *[OCEL: Automated remediation]*

Attribute	Type	Constraint	Description
id	UUID	PK	Primary key
tenant_id	UUID	FK→Tenant	Parent tenant
package_id	UUID	FK→Package	Parent package
key	VARCHAR(100)	NOT NULL	Unique key
name	VARCHAR(255)	NOT NULL	Skill name
description	TEXT	NULL	Description
sensor_id	UUID	FK→Sensor	Detection sensor
action_flow_id	UUID	FK→ActionFlow	Automation flow
status	ENUM	NOT NULL	draft|active|inactive
signal_count	INTEGER	COMPUTED	Signal count
last_evaluated_at	TIMESTAMPTZ	NULL	Last evaluation
created_by	UUID	FK→User	Creator
created_at	TIMESTAMPTZ	NOT NULL	Creation timestamp

Relationships: N:1 → Package | N:1 → Sensor | N:1 → ActionFlow | 1:N → Signal

Relationship Matrix

This section summarizes the key relationships between entities across ontological layers.

Source Layer	Source Entity	Relationship	Target Entity	Target Layer
Existence	Tenant	1:N	Organization	Existence
Existence	Tenant	1:N	User	Identity
Existence	Tenant	1:N	DataPool	Data Infrastructure
Data Infrastructure	DataPool	1:N	DataModel	Data Infrastructure
Data Infrastructure	DataModel	1:N	Perspective	OCPM Core
OCPM Core	Perspective	1:N	ObjectType	OCPM Core
OCPM Core	Perspective	1:N	EventType	OCPM Core
OCPM Core	ObjectType	1:N	Object	OCPM Core
OCPM Core	Object	N:M	Event	Case-Centric
Case-Centric	Case	1:N	Event	Case-Centric
Case-Centric	Variant	1:N	Case	Case-Centric
Semantic	KnowledgeModel	1:N	KPI	Semantic
Semantic	KnowledgeModel	1:N	Sensor	Automation
Automation	Sensor	1:N	Signal	Automation
Automation	Signal	1:1	Task	Automation
Studio	Package	1:N	View	Studio
Studio	Package	1:N	ActionFlow	Automation

PM4Py Integration Guide

This section provides mapping between the ERD entities and PM4Py library objects, enabling seamless integration with the Python process mining ecosystem.

ERD Entity	PM4Py Class	Usage
Event	`pm4py.objects.log.obj.Event`	Single event occurrence
Case	`pm4py.objects.log.obj.Trace`	Case/trace container
Variant	`pm4py.statistics.variants`	Variant analysis
Object (OCPM)	`pm4py.objects.ocel.obj.OCEL.objects`	OCEL object instances
ObjectType	`pm4py.objects.ocel.obj.OCEL.object_types`	OCEL object type definitions
EventType	`pm4py.objects.ocel.obj.OCEL.event_types`	OCEL event type definitions
EventObjectRelationship	`pm4py.objects.ocel.obj.OCEL.relations`	E2O relationships
ObjectRelationshipInstance	`pm4py.objects.ocel.obj.OCEL.o2o`	O2O relationships
DataModel	`pm4py.read_ocel / pm4py.read_xes`	Log import configuration
Perspective	`OCEL filtering`	Perspective-based views

OCEL 2.0 Attribute Mapping

OCEL 2.0 Attribute	ERD Column	Table
`ocel:eid`	`id`	Event
`ocel:timestamp`	`timestamp`	Event
`ocel:activity`	`activity_name`	Event
`ocel:oid`	`id`	Object
`ocel:type`	`object_type_id → name`	Object → ObjectType
`ocel:ovmap`	`attributes (JSONB)`	Object
`ocel:evmap`	`attributes (JSONB)`	Event
`ocel:e2o`	`event_id, object_id, qualifier`	EventObjectRelationship
`ocel:o2o`	`source_object_id, target_object_id`	ObjectRelationshipInstance